

Florentin C. Jaeger

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Research Interests Plant morphological and physiological ecology, soil ecology, fungi, global change related disturbances, plant phenotypic plasticity, remote sensing, molecular ecology, land management

EDUCATION

- Université du Québec à Montréal**, Montréal, QC 01/03/2018—05/12/2024
Ph.D., Biology
Rooting for resilience: Unveiling absorptive fine root traits and fungal dynamics in young temperate trees under high and low growing season water availability
Advisors: Christian Messier & I. Tanya Handa
- University of Freiburg**, Freiburg im Breisgau, Germany 01/10/2014—25/08/2017
M.Sc., Forest Science
Effects of mixing *Fagus sylvatica* and *Abies alba* on belowground complementarity at a submontane site of the Buntsandstein region of Baden-Württemberg
- University of Applied Forest Science Rottenburg**, Germany 01/10/2009—20/05/2014
B.Sc., Forest management, Spec. in Forest Economics

RESEARCH EXPERIENCE

- Cornell University**, Ithaca, NY 16/01/2025—15/01/2026
Postdoc, Tree species carbon partitioning and reforestation using self-burying seed carriers
Rooted in Nature: Bioinspired Design of Sustainable Seeding Methods to Improve Forest Regeneration
Whole tree carbon partitioning and root anatomical and exudation strategies of six common North American species
Advisors: Taryn L. Bauerle & Marc Goebel

TEACHING EXPERIENCE

- Teaching Assistant**
PLSCI 1115: The Nature of Plants (Guest Lecturer) May 2025
Cornell University
Lead instructor: Taryn L. Bauerle
- BIA3010 - Méthodologie en écologie 12/08/2024—16/08/2024
Université du Québec à Montréal

Lead instructor: I. Tanya Handa

Instructor

CEF workshop

20/09/2024

Centre d'étude de la forêt

Zotero & Obsidian for literature management and notetaking

R Workshop Series

Spring 2021

Québec Centre for Biodiversity Science

Linear and generalized linear mixed models (LMM and GLMM)

PUBLICATIONS

Published **Jaeger, F.C.**, Handa, I.T., Paquette, A. et al. Young temperate tree species show different fine root acclimation capacity to growing season water availability. *Plant Soil* 496, 485–504 (2024). <https://doi.org/10.1007/s11104-023-06377-w>

This study showed that six temperate tree species differed markedly in how they adjusted absorptive fine-root biomass and rooting depth to contrasting growing-season water availability. It highlighted that root acclimation is strongly species-specific and that both biomass investment and vertical root distribution are important components of tree responses to altered water regimes.

Jaeger, F.C., Messier, C., Beyer, F. et al. Young temperate tree species show fine root trait acclimation to differences in water availability. *Plant Soil* 518, 1011–1034 (2026). <https://doi.org/10.1007/s11104-025-08047-5>

This study demonstrated that fine-root architectural and morphological traits respond differently to water availability across tree species, with root branching density emerging as a particularly important drought-response trait. It further showed that early- and late-successional species differ in their fine-root plasticity, helping identify belowground traits that may contribute to resilience under changing water availability.

Sanaei, A., Saxena, S., Mueller, K.E., McCormack, M.L., Schuldt, B., Laughlin, D.C., Rosado, B.H.P., Freschet, G.T., Addo-Danso, S.D., Barry, K.E., Bergmann, J., Eisenhauer, N., **Jaeger, F.C.**, Poorter, H., Olde Venterink, H., De Vries, J., Weemstra, M., Mommer, L. and Weigelt, A. (2026), On the relationship between root economics and plant hydraulic traits. *New Phytol*, 251: 3289-3302. <https://doi.org/10.1111/nph.71442>

Manuscripts In Preparation **F. C. Jaeger**, C. Messier, Z. Wang, W. C. Parker, C. Fahey, P. M. Antunes, I. T. Handa, S. W. Kembel. Root fungi and absorptive fine root traits under low and high water availability. (to be submitted to the *New Phytologist* this fall)

E. Gagnon-Koudji, **F. C. Jaeger**, and I. T. Handa. Effects of higher and lower water availability on soil springtail communities and their interaction with tree litter and roots. (to be submitted to *Frontiers in Ecology and the Environment* this fall)

M. Mohr, **F. C. Jaeger**, J. A. Schwarz, J. Bauhus, F. Beyer. Drought forces similar root trait acclimations in European beech and silver fir irrespective of their mixed or monospecific neighborhood. (to be submitted to *Forest Ecology and Management* this fall)

Monique Weemstra, Kathryn E. Barry, **F. C. Jaeger**, Team FunDry, Alexandra Weigelt, Liesje Mommer. Greater biomass allocation to roots is not a pervasive answer to aridity globally. (Target journal: *Functional Ecology*)

Semina Yi, Yaoye Hong, **F. C. Jaeger**, Marco Freschi, Yaobin Yang, Ziyun Zhang, Hajun Lee, Kenichi Soga, Ehren Moler, Teng Zhang, Taryn L. Bauerle, Shu Yang, Lining Yao. Hygrochastic Seedpods: Engineering Microclimates for Ecological Restoration. (Target journal: *Nature Ecology and Evolution*, submission this fall)

Contributed Research Presentations **F. C. Jaeger**, C. Messier, Z. Wang, W. C. Parker, C. Fahey, P. M. Antunes, I. T. Handa, S. W. Kembel. *Root traits and fungi shift under changes in water availability*. 17^e Colloque du Centre d'étude de la forêt 2024. Talk. May 2024.

F. C. Jaeger, I. T. Handa, W. C. Parker, A. Paquette, C. Messier. *Absorptive fine roots of temperate tree species show acclimation capacity to changes in growing season water availability*. North American Forest Ecology Workshop 2022. Talk. June 2022.

F. C. Jaeger, J. Schwarz, and J. Bauhus. *Effects of mixing Fagus Sylvatica and Abies alba on below-ground complementarity*. North American Forest Soils Conference 2018. Poster. 2018.

F. C. Jaeger, U. Schroeder-Caldas, and A. Petkau. *What are the impacts of the new Brazilian legislation on forests from 2001 on the behavior of the local forest owners in regard to the forest treatment? Are there any changes visible because of the revision in 2012?* 7th German-Brazilian Symposium for Sustainable Development 2015. Talk. October 2015.

Peer review Perspectives in Plant Ecology, Evolution and Systematics — 29/20/2025
Invited peer reviewer

Plant and Soil — Invited peer reviewer 04/01/2025

Patent application & innovation **F. C. Jaeger** [and co-inventors]. Multi-phase delivery system for self-burying engineered seed carrier. Patent application, U Cal No. BK-2026-107; MN No. 407869-0251(2026). Bio-inspired technology combining parachute-assisted landing with water-responsive self-burial for seed delivery and ecological restoration.

Funding Fellowships & Awards Cornell University Faculty Advising, Teaching and Mentorship Award — Innovation category summer 2025

DAAD UNIBRAL Scholarship for Research in Brazil (€9,600) 01/01/2013–23/12/2013

GIZ–FH Programme “Foreign Practical Semester”
(€4,800)

01/10/2011–29/01/2012

Mentoring	Bailey xxxx, M.P.S. student at Cornell University Project title: <i>Seed germination of four common Northeastern American tree species and seedling root interactions</i> Role: Project mentor (with Taryn L. Bauerle)	2024—2025
	Jérémi xxxx, M.Sc. student at Université du Québec à Montréal Project title: <i>La communauté des Collemboles de la forêt urbaine de Montréal de part son historique, sa gestion et sa complexité</i> Role: Provided informal peer support in R coding and data analysis; feedback on writing and literature organization (Zotero workflows). (Project mentor: I. Tanya Handa)	2019—2024
	Roberto xxxx, M.Sc. student at UQAM Project title: <i>Influence of urban forest complexity on the soil macrofauna communities and associated ecosystem services</i> Role: Provided informal peer mentoring in research organization, Zotero use, and thesis writing support. (Project mentor: I. Tanya Handa)	2020—2024
Science communication	Expanding Your Horizons (EYH) STEM Conference Cornell University — Co-organized hands-on plant physiology outreach for 7th–10th grade students.	05/04/2025
Service	Member of the International Forestry Students Association (IFSA) and organizer of regional and international meetings in Rottenburg and Freiburg	2009—2017

SKILLS

Field & Greenhouse Methods

- Experimental site establishment, management, and monitoring
- Applied wildlife management (game monitoring and hunting certification)
- Tree seed stratification, germination, and seedling establishment
- Soil and root coring, root scanning (WinRHIZO) and functional sorting, root trait analysis (SRL, RTD, branching density, etc.)
- Root exudate sampling (Phillips method) and non-structural carbohydrate (NSC) analysis
- Segmentation of CT scans (manual & deep learning in ORS Dragonfly 3D World)
- Design and execution of Seedpod experiments in the greenhouse

Molecular & Bioinformatics Techniques

- DNA extraction and purification (QIAGEN kits), PCR (thermocycler), gel electrophoresis
- Fungal DNA metabarcoding (ITS2 region, Illumina MiSeq)
- Bioinformatics pipelines: Cutadapt, DADA2, UNITE database taxonomy assignment
- Fungal guild functional assignment (FUNGuild)

Statistical & Computational Skills

- R for statistical computing (tidyverse, lme4, vegan, car, emmeans, piecewiseSEM, etc.)
- Mixed models (LMM/GLMM), multivariate statistics (PERMANOVA, NMDS, PCA, PCoA), structural equation modeling
- Bootstrapping and relative importance metrics (relaimpo)
- Longitudinal/time-series modeling in R fitting nlme::gls/lme with AR(1) autocorrelation; geepack::geeglm GEE for binomial emergence; emmeans for EMMs/contrasts and growth-rate trends
- Reproducible workflows with R Quarto (HTML, PDF, revealjs) and publication on GitHub
- Bash scripting and command-line use (Ubuntu/Linux)

Software & Tools

- Zotero (with advanced extensions), Obsidian for knowledge management
- Microsoft Office 365, LibreOffice
- Arduino sensor setup and data logging

Languages

- English (fluent), German (fluent), French (beginner), Portuguese (beginner), Spanish (beginner)

Other Professional Skills

- Chainsaw and field safety certification
- Qualitative interviews for socio-economic research
- Conference presentations (talks and posters across North America and Europe)
- Hunting license (Germany)
- Drivers license (Germany & New York State)